20260416 BEL 2

14 A spare-parts store uses a reorder-point logic together with target inventory correction law.

a) Given

$$\bar{d} = 18 \frac{\text{units}}{\text{day}}, \qquad L = 4 \text{ days},$$

$$SS = 12 \text{ units},$$

compute the reorder point

$$R = \bar{d} L + SS$$

$\bar{d}$: Mean daily demand
L : lead time
SS : Safety stock

$\Rightarrow$ Calculation

$$R = 18 \frac{\text{units}}{\cancel{\text{day}}} \cdot 4\, \cancel{\text{days}} + 12 \text{ units}$$

$$R = 18 \cdot 4 + 12 \text{ units} = 84 \text{ units}$$

$\Rightarrow$ When the inventory drops to 84 units, we need to re-order the parts!

b) Let the replenishment rule be:

$$q(t) = 0.5[I^*(t) - I(t)]$$

and the inventory balance:

$$\dot{I}(t) = q(t) - d(t)$$

$\Rightarrow$ derive the diff eq. for $I(t)$!

⇒ We just substitute $q(t) = 0.5\,[I^*(t) - I(t)]$ into the inventory balance equation:

$$\dot{I}(t) = 0.5\,[I^*(t) - I(t)] - d(t)$$

$$\dot{I}(t) = 0.5\,I^*(t) - 0.5\,I(t) - d(t)$$

$$\boxed{\dot{I}(t) + 0.5\,I(t) = 0.5\,I^*(t) - d(t)}$$

⇒ We see that this is a 1st-order system, and we can express it differently to see the time constant τ.

$$\boxed{\begin{gathered}\underline{2}\,\dot{I}(t) + I(t) = I^*(t) - 2\,d(t) \\ \tau = 2 \text{ days}\end{gathered}}$$

c) For the case $d(t) = 0$, determine the transfer function (TF).

$$\frac{I(s)}{I^*(s)}$$

⇒ We follow the steps:

1) Set initial conditions to zero ✓
2) Swap derivatives for s (take it out of time domain)

$$\Rightarrow \quad 2s\,I(s) + I(s) = I^*(s) \quad ✓$$

3) Rearrange:

$$(2s + 1)\,I(s) = I^*(s)$$

$$\boxed{\frac{I(s)}{I^*(s)} = \frac{1}{2s + 1}} \quad ✓$$

Analysis of system:

- It has a static gain of 1 $\Rightarrow$ set $s=0$ ✓
- It has a τ of 2 days ✓
- It has a pole $p = -\frac{1}{2}$ $\Rightarrow$ stable ✓

$\boxed{16}$ A base-stock inventory model under demand disturbance is

$$\dot{I}(t) + 2I(t) = -d(t), \qquad I(0) = 0$$

a) Determine the transfer function

$$G(s) = \frac{I(s)}{D(s)}$$

$\Rightarrow$ Let's re-express the 1st-order diff eq:

$$\frac{1}{2}\dot{I}(t) + I(t) = -\frac{1}{2}d(t)$$

$\Rightarrow$ steps 1-3

1) set initial conditions to zero
2) Swap derivatives for s

$$\frac{1}{2}sI(s) + I(s) = -\frac{1}{2}D(s)$$

3) Rearrange

$$\left[\frac{1}{2}s + 1\right] I(s) = -\frac{1}{2}D(s)$$

$$\Rightarrow \boxed{G(s) = \frac{I(s)}{D(s)} = \frac{-0.5}{0.5s + 1}}$$ ✓

$$\tau = 0.5$$

b) For the demand step

$$d(t) = 6\,u(t),$$

compute the inventory response $I(t)$.

$\Rightarrow$ First convert the step function to Laplace:

$$D(s) = \frac{6}{s}$$

$\Rightarrow$ Set up the resulting system:

$$I(s) = G(s)\,D(s) = \left(\frac{-0.5}{0.5s+1}\right)\frac{6}{s}$$

$$= \frac{-3}{s(0.5s+1)} = \frac{-6}{s(s+2)}$$

get the denom. like $s(s+a)$

$\Rightarrow$ Apply the inverse Laplace transform

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s+a)}\right\} = \frac{1}{a}(1-e^{-at})\,u(t)$$

$$\Rightarrow \quad \mathcal{L}^{-1}\left\{\frac{-6}{s(s+2)}\right\} = -\frac{6}{2}(1-e^{-2t})\,u(t)$$

$$\boxed{I(t) = -3(1-e^{-2t})\,u(t)} \checkmark$$

c) Determine the long-term inventory deviation and explain its logistics meaning.

$\Rightarrow$ Let's take the limit as $t \to \infty$

$$\lim_{t\to\infty} -3(1-e^{-2t})\,u(t)$$

$$= -3(1-0)\cdot 1 = -3 \checkmark$$

$\Rightarrow$ The long-term value for inventory deviation if we see an instantaneous jump in demand of 6 units/day, our inventory will settle to

$$\boxed{\text{initial value } - 3}$$

[19] A CONWIP-controlled production line uses

$r(t) = c(t) + 0.25[W^*(t) - W(t)]$,

$\dot{W}(t) \quad r(t) - c(t)$

a) Derive the closed WIP equation

$$\dot{W}(t) + 0.25\,W(t) = 0.25\,W^*(t)$$

$\Rightarrow$ We substitute in the release rule $r(t)$

$$\Rightarrow \quad \dot{W}(t) = [\cancel{c(t)} + 0.25(W^*(t) - W(t))] - \cancel{c(t)}$$

$$\boxed{\dot{W}(t) + 0.25\,W(t) = 0.25\,W^*(t)} \quad ✓$$

b) Determine the TF

$$G(s) = \frac{W(s)}{W^*(s)}$$

$\Rightarrow$ Steps:

1) IC to zero ✓
2) Swap derivatives

$$s\,W(s) + 0.25\,W(s) = 0.25\,W^*(s)$$

3) Rearrange

$$\boxed{\frac{W(s)}{W^*(s)} = \frac{0.25}{s + 0.25}}$$

$\Rightarrow$ Time constant form:

$$\frac{1}{4s + 1} \qquad \tau = 4 \qquad p_1 = -0.25$$

c) For the target step

$$W^*(t) = 120\,u(t), \qquad W(0) = 0$$

compute the WIP response!

$$\Rightarrow \quad W(s) = G(s)\,W^*(s)$$

$$W(s) = \left(\frac{0.25}{s+0.25}\right)\frac{120}{s}$$

$$W(s) = \frac{30}{s(s+0.25)} \quad ✓$$

$\Rightarrow$ We apply the inverse Laplace transform:

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s+a)}\right\} = \frac{1}{a}\left(1-e^{-at}\right)u(t)$$

$$\Rightarrow \quad \mathcal{L}^{-1}\left\{\frac{30}{s(s+0.25)}\right\} = \frac{30}{0.25}\left(1-e^{-0.25t}\right)u(t)$$

$$\boxed{W(t) = 120\left(1-e^{-t/4}\right)u(t)} \quad ✓$$

d) If the steady-state throughput at the target WIP is

$$TH = 30 \ \frac{\text{jobs}}{\text{day}}$$

use Little's law to determine the corresponding cycle time CT.

$$WIP \approx TH \times CT$$

$$CT \approx \frac{WIP}{TH} \approx \frac{120 \text{ jobs}}{30 \ \frac{\text{jobs}}{\text{day}}}$$

$$\boxed{CT \approx \frac{120}{30}\text{ days} = 4 \text{ days}} \quad ✓$$